

# The AI nanobody design pipeline, and where each model sits

## The pipeline

Going from "we want a binder against HER2" to "here are 20 sequences worth ordering" is five stages. Each one is a filter: it takes a larger, cheaper set of candidates and hands the next stage a smaller, better-characterised one. The expensive structure-based tools sit late precisely because they cost too much per candidate to point at a naive library.

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|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| (0) TARGET DEFINITION | HER2 seq/structure -> chosen epitope ("hotspot")                                                                                             |
| (1) GENERATION        | (a) wet lab: immunize alpaca, pan, NGS <-- this test<br>(b) de novo: generate backbones in silico<br>-> $10^2..10^6$ candidate VHH sequences |
| (2) CHEAP RANKING     | score every candidate, sequence only, no folding<br>-> ranked list; keep top ~1-5% <-- Tasks 1 and 2                                         |
| (3) STRUCTURAL TRIAGE | fold survivors, dock, score interface<br>-> complex models, affinity estimates                                                               |
| (4) DEVELOPABILITY    | stability, aggregation, immunogenicity -> ~10-50 seqs                                                                                        |
| (5) ORDER AND TEST    | express, purify, SPR/BLI -> real $K_D$<br>`-----> feeds back into (2)                                                                        |

Two things about this shape matter more than any individual model.

**The funnel is economic, not intellectual.** Stage 2 is cheap and inaccurate; stage 3 is expensive and better; stage 5 is the only ground truth. You place a model by asking what it costs per candidate and how much it narrows the field.

**The loop closes at stage 2.** Measured affinities from stage 5 are the training data for the next round's ranker. That is the stage this coding test is about: our model consumes a noisy enrichment proxy ( `target_IE_score` ) rather than  $K_D$ , which is exactly the data a first round actually produces.

## Where each model fits

Grouped by the job they do, not by lab or architecture.

### Structure prediction — "what shape is this, and does it dock?"

**AlphaFold-Multimer** (2021). *In*: two or more sequences. *Out*: a structure for the complex, with per-interface confidence. The workhorse for asking whether a designed VHH and its target form a plausible interface. Stage 3.

**AlphaFold3** (2024). *In*: sequences plus, unlike its predecessors, nucleic acids, ligands and covalent modifications. *Out*: a joint structure for the whole assembly. Same stage-3 job as Multimer, broader input vocabulary.

**ESMFold2**. *In*: a single sequence, no MSA. *Out*: a predicted structure. The trade is speed for accuracy: MSA-free folding is fast enough to run over thousands of candidates, which makes it the tool for a *first* structural pass at the top of stage 3 where AlphaFold would be unaffordable.

**Rosetta** (Nature Methods 2020 for the modern suite). *In*: a structure, plus a protocol. *Out*: energies, minimised structures, docked poses, designed sequences. Physics-based rather than learned, and the oldest thing on this list. Still the default for stage-4 questions — relax this interface, estimate a  $\Delta\Delta G$  for this mutation — and for scoring the outputs of the generative models below.

### Representation — "turn a sequence into numbers"

**ESM / ESMC** (the family used in Task 1). *In*: a sequence. *Out*: a per-residue embedding, and a likelihood for each position. Two distinct uses: as a frozen feature extractor for a small supervised head (what Task 1 does), and as a zero-shot fitness proxy, where a low-likelihood mutation is a guess at "this protein is less viable". Stage 2, and the cheapest thing here — no structure, one forward pass.

### Generative design — "invent a binder that wasn't in any library"

**RFdiffusion** (2023; antibody-specific version 2025). *In*: a target structure and an epitope to hit. *Out*: novel backbones designed to bind there, with sequences assigned by a downstream design step. This is the stage-1(b)

alternative to immunizing an animal. The 2025 antibody work extends it to designing CDR loops on a fixed framework, which is the nanobody-relevant case.

**BoltzGen** (2025). *In*: a target specification. *Out*: generated binder designs. Occupies the same stage-1(b) slot as RFdiffusion.

**Germinal** (Nature Biotechnology 2026). *In*: a target and epitope. *Out*: de novo antibody/nanobody designs. Presented as an end-to-end pipeline rather than a single model, i.e. generation plus filtering in one workflow — stages 1(b) through 4.

### Affinity prediction — "how tightly, not just whether"

**Boltz-2** (2025). *In*: a complex specification. *Out*: a structure **and** a binding affinity estimate. Notable because predicting affinity rather than just geometry is what stage 3 actually needs: a ranked shortlist, not a pile of plausible poses.

**BoltzProt-1** (2026). *In*: protein sequence/structure input. *Out*: a general-purpose protein model from the Boltz line; no separate public repository, released via the `boltzgen` codebase and an API.

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## Two honest caveats

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**On the newest three.** The specific claims above for **Germinal**, **BoltzProt-1** and **ESMFold2** are the least certain — they are recent enough that I would verify the input/output description against the linked papers before defending any of it in detail. The stage each occupies is the part I am confident about; the exact interface is worth a five-minute check.

**On what none of them do.** Every model here reasons about structure, sequence plausibility or affinity. None predicts what actually kills nanobody programmes: expression yield, aggregation on storage, polyreactivity, or immunogenicity in a human patient. Stage 4 is still mostly heuristics and assay data, and stage 5 is still the only thing that settles an argument.